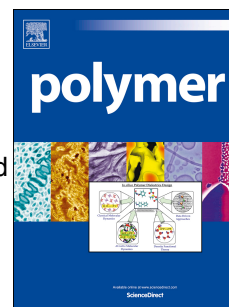


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Confinement effect on spatio-temporal growth of spherulites from cellulose/ionic liquid solutions

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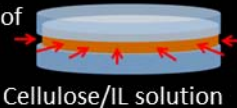
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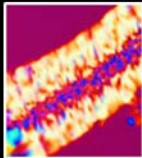
Effect of confinement

20% RH 40°C

Diffusion of
water

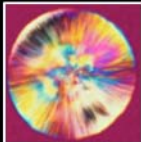


Crystallization of cellulose

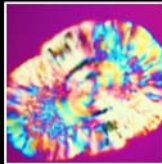


shish-kebab

spherulites



terraces



elongated

Confinement effect on spatio-temporal growth of spherulites from cellulose/ionic liquid solutions

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Abstract

In this study, we investigate the growth kinetics and through-the width-distribution of polycrystalline morphologies developed in the confined microcrystalline cellulose (MCC)/1-allyl-3-methylimidazolium chloride (AmimCl) solution films at controlled temperature and humidity. Being an anti-solvent, the diffusion of moisture from the edges to the interior of the cellulose film causes the aggregation and rearrangement of cellulose molecules. When the confined films of thickness varying between 100 μm and 850 μm are incubated for over 25 days, different polycrystalline morphologies evolve. While a skin-transition-core morphological distribution is observed in the films of thickness $\geq 700 \mu\text{m}$, interesting morphologies such as cylindrites, *shish-kebab*, deformed spherulites are observed in confined films of thickness $\leq 500 \mu\text{m}$. The formation of concentric rings called terraces is also observed on the spherulites in highly confined films, signaling two dimensional (2-D) growth. The growth kinetics is observed to follow the classical Avrami theory. The Avrami exponent value, n , in the highly confined film (100 μm) is found to

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be close to 2, indicating 2-D growth.

Keywords: Spherulites, *shish-kebab*, confinement, cellulose/ionic liquid solution, Avrami kinetics, terraces

1. Introduction

Self-assembly and crystallization of biopolymers is an area of great interest. Spherulite is the most common morphology observed during crystallization of polymers. They are polycrystalline aggregates grown in radial direction from a nucleation center, from melts or concentrated solutions. The formation of spherulitic morphologies is found in concentrated solutions of various biomacromolecules, such as DNA, silk, chitin, chitosan, cellulose, etc. [1–11] In this context, even though cellulose is the most abundant, inexhaustible biopolymer, the formation of polycrystalline structures composed of cellulose molecules has received very little attention. This is due to the difficulty in dissolving cellulose at sufficiently high concentrations required for the formation of spherulites.

Recently, a new class of solvents, namely, ‘ionic liquids’ (IL) is found to dissolve cellulose in high concentrations [12–15]. In spite of several advantages of ILs, such as high vapor pressure, excellent thermal stability, easy recyclability etc., most of the ILs are very hygroscopic. The hygroscopic nature of IL hinders the dissolution of cellulose by decreasing the solvent quality. The other side of the coin is the utilization and control of this intrinsic hygroscopic nature of ILs to induce self-assembly of cellulose molecules. This is due to the freeing up of hydrogen bonding sites of the cellulose molecules arising due to preferred affinity of ILs towards water leading to the local

aggregation and rearrangement of cellulose chains. Recently, we studied the formation and transition of cellulose/IL solutions into various phases – liquid crystalline phase to cellulose spherulites – when anti-solvent (water) is added [16]. There are a few reports on the spherulite formation in both neat and nanofiller-loaded cellulose/IL solutions by the diffusion of anti-solvent (water) [17–19]. In these studies, the moisture diffuses into the cellulose film coated on a solid substrate resulting in the nucleation of cellulose spherulites. This leads to the formation of standard spherulites of different types, such as positive/negative or banded/non-banded with time.

This study involved the investigation of the effect of confinement on the crystallization behavior of cellulose in microcrystalline cellulose (MCC)/1-allyl-3-methylimidazolium chloride (AmimCl) solutions via diffusion of anti-solvent (moisture) and the distribution of morphologies thereof. Polymer crystallization under confined conditions is of great importance. Various processes and parameters such as kinetics of crystallization and crystalline morphology are greatly altered by the confinement, which in turn influence the physical properties of the resulting material [20]. However, as the confinement thickness increases beyond ~ 100 nm, the standard spherulites as observed in the bulk crystallization is observed [20]. In this study, by confining the cellulose film between substrates, it is possible to direct the diffusion of moisture and achieve control over the rate of diffusion by altering the confinement thickness. The confinement method naturally creates areas with different degrees of anisotropy in the sample, in terms of the driving force for crystallization. This leads to the formation of different types of polycrystalline morphologies during the crystallization of cellulose, even when the

confinement thickness is of few hundreds of micrometers, rather than the standard spherulites formed in the films coated over the substrate [17]. In our study, we observed the formation of polycrystalline morphologies such as skin-transition-core morphological distribution, cylindrites, shish-kebabs, bridging between spherulites and deformed spherulites depending on the confinement thickness. The skin-core morphological distribution and the growth of cylindrites and *shish-kebab* are characteristic of shear-induced crystallization, which greatly enhance the properties of the material. Here, the formation of such morphologies in the absence of shear is hypothesized to be due to the directional diffusion of moisture from the air-film interface to the interior of the film, causing the crystallization of cellulose molecules along the diffusion path and further crystallization of the core region or the *shish*, which acts as the nucleation site for the lateral growth of *kebabs*. Furthermore, we study the kinetics of crystallization and influence of confinement by the classical Avrami theory.

2. Experimental Section

2.1. Materials used

AmimCl was purchased from Sigma Aldrich and MCC with degrees of polymerization ~ 350 , purchased from Alfa Aesar was used as the cellulose source.

2.2. Preparation of MCC/AmimCl solution

20 wt% MCC/AmimCl solution was used for preparation of cellulose spherulites. A known quantity AmimCl, about 5 g, was transferred to a

Schlenk flask and freeze dried overnight to remove traces of water. MCC was dried overnight in an oven at 80 °C to remove moisture. The freeze dried AmimCl was then melted at 100 °C in a constant temperature oil bath with continuous stirring under high vacuum. A known quantity of dried MCC was added to the molten AmimCl, under a dry N₂ environment, to avoid absorption of moisture. The mixture was then stirred continuously for about 8 hours under high vacuum of $\sim 10^{-4}$ mbar at 80 °C until the complete dissolution of cellulose. Stirring under vacuum was observed to help in the removal of entrapped air bubbles, if any. The process was monitored by bright field and polarization optical microscopy to ensure complete dissolution of cellulose. The moisture content in the prepared samples were measured using Karl-Fischer titration and was found to be < 1 wt%. The prepared MCC/AmimCl solution was kept under quiescent condition in a vacuum desiccator until further experiments.

2.3. Experimental setup

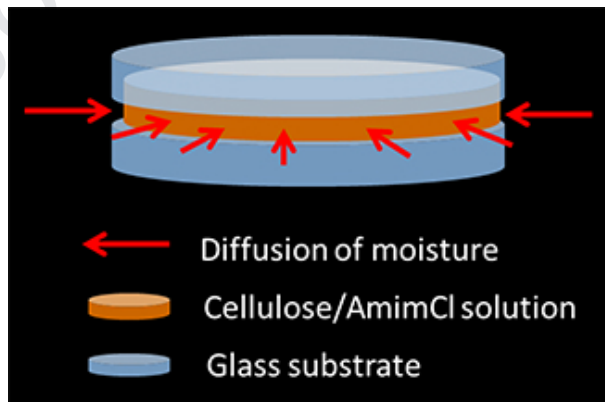


Figure 1: Schematic showing the experimental setup

Figure 1 shows the schematic of experimental setup for the controlled diffusion of anti-solvent (water) into the MCC/AmimCl film. For this purpose, the prepared MCC/AmimCl solution was heated under high vacuum with stirring at 80 °C for about 15 minutes to reduce the viscosity of the solution. Two clean glass substrates were also preheated to 80 °C on a heating plate for about 15 minutes. A known volume of MCC/AmimCl solution was placed exactly at the center of one of the glass substrates and the second glass substrate was placed on the top of it to achieve a cellulose film of uniform thickness, as shown in Figure 1. This confined cellulose film is then left undisturbed for about 15 minutes at 80 °C, to remove any thermal history and to allow the solution to spread uniformly between the glass substrates. By varying the volume of the cellulose/IL solution, the film thickness was systematically varied from 100 μm to 850 μm . The film thickness was measured using a digital micrometer to an accuracy of $\pm 1.27 \mu\text{m}$. The confined cellulose film was maintained at a constant temperature of 40 °C using a hot air oven and 20% relative humidity (RH) by means of a desiccator containing self-indicating silica gel. The temperature and humidity were constantly monitored by placing suitable sensors inside the desiccator. It should be noted that the temperature and humidity conditions of the desiccator were maintained at the experimental conditions for 24 hours prior to placing the cellulose film. The open edges of the film will enable the diffusion of ambient moisture (anti-solvent) into the MCC/AmimCl solution film and facilitate the growth of cellulose spherulites. Since the solution is of high viscosity ($\sim 2000 \text{ Pa}\cdot\text{s}$ at 25 °C), no appreciable change in the thickness of the film was noticed during the course of experiments.

2.4. Polarized optical microscopy (POM)

The confined cellulose film was observed for the formation of cellulose spherulites at specific intervals. Leica DMI3000B inverted microscope was used for recording images using $5\times$ objective. The higher magnification images ($10\times$ objective and above) were acquired with Nikon Eclipse LV100ND microscope equipped with crossed polarizers. To enhance the contrast further, a 530 nm tint plate (1λ) (P-CL Nikon) was placed diagonally between the cross polarizers resulting in a magenta background.

3. Results and discussion

When the MCC/AmimCl solution films confined between glass surfaces are incubated in ambient air at a $40\text{ }^{\circ}\text{C}$ and 20% relative humidity (RH), water (anti-solvent for cellulose) diffuses from the ambient air through the air-film interface to the interior of the film. This leads to the aggregation of cellulose and generation of nuclei for the formation of polycrystalline morphologies. The evolution of morphology depends on the distance from the air-film interface, thickness of the film and incubation time. During the incubation, the evolution of the microstructure is monitored every 6 hours. Within 3 days of incubation, depending upon the nature of cellulose crystallization across the film, it is possible to perceive the presence of two distinct zones, as depicted schematically in Figure 2. Initially, the number of nuclei formed near the air-film interface is high due to the diffusion of large number of water molecules into the film. This leads to the formation of small spherulites ($\sim 10\text{ }\mu\text{m}$ diameter) and/or superimposed spherulitic structures in the annular region designated as ‘zone I’ in Figure 2 in the immediate vicinity

of the air-film interface. Whereas, the lower concentration of water molecules in the interior of the cellulose film due to the mass transfer resistance offered by the film, leads to the formation of few individual spherulites, however, of much larger diameter ($\sim 100 \mu\text{m}$). This zone is schematically represented in Figure 2 as 'zone II'. A representative POM image showing growth of 'zone I' in $500 \mu\text{m}$ after 3 days is given in Figure S 1. The boundary between zone I and zone II remains dynamic as the microstructure evolves due to continuous diffusion of water molecules into the film. The rationale behind the classification of the across-width region as different zones is to aid the discussions on the evolution and distribution of different morphologies in the confined MCC/AmimCl solution films.

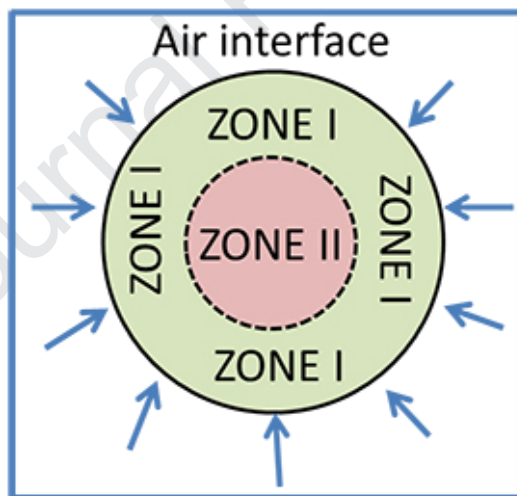


Figure 2: Schematic showing the across-the film distribution of morphologies as different zones with respect to the air-film interface. The solid circle denotes the air-film interface and the dashed circle shows the dynamic boundary between zone I and zone II.

We investigate the evolution of microstructure at different confinement

thickness – 100, 200, 300, 500, 700 and 850 μm . Based on the growth pattern and morphologies observed, we categorize the microstructural transition as: (a) early stage growth (within 3 days of incubation period) (b) intermediate stage (4 – 10 days of incubation), and (c) terminal growth stage (11 – 25 days of incubation). The presence/absence of the zone I and zone II in different stages of growth (morphological evolution) is consolidated in Table S1 (ESI).

3.1. Early stage growth (within 3 days of incubation period)

Figure 3 shows POM images that demonstrate the effect of confinement on the evolution of morphologies in the early growth stage.

The first evident observation is the distinguishable difference in the size and number of spherulites that form in zone I depending on the confinement thickness. Within 24 hours of incubation, spherulites of $\sim 50 - 70 \mu\text{m}$ diameter, as shown in Figure 3 (a) and (b), are observed to form in the cellulose/IL films of thickness $\leq 500 \mu\text{m}$. On the 3rd day of incubation, the average diameter of spherulites is found to increase to $100 - 150 \mu\text{m}$, as shown in Figure 3 (d) and (e). Therefore, in the early incubation stage, the diameter of the spherulites increases with increase in the thickness of the cellulose/IL film (i.e., decrease in the confinement). This is shown in Figure S 2 (a) (ESI).

The span of zone I from the edge of the film towards the center (i.e., the width of zone I), quantified after 3 days of incubation, is found to increase with increase in the film thickness, as shown in Figure S 2 (b) (ESI). Interestingly, in the thicker films (i.e., 700 μm and 850 μm films), the growth of several hundreds of tiny spherulites of size $\sim 10 \mu\text{m}$ (Figure 3 (c)) is ob-

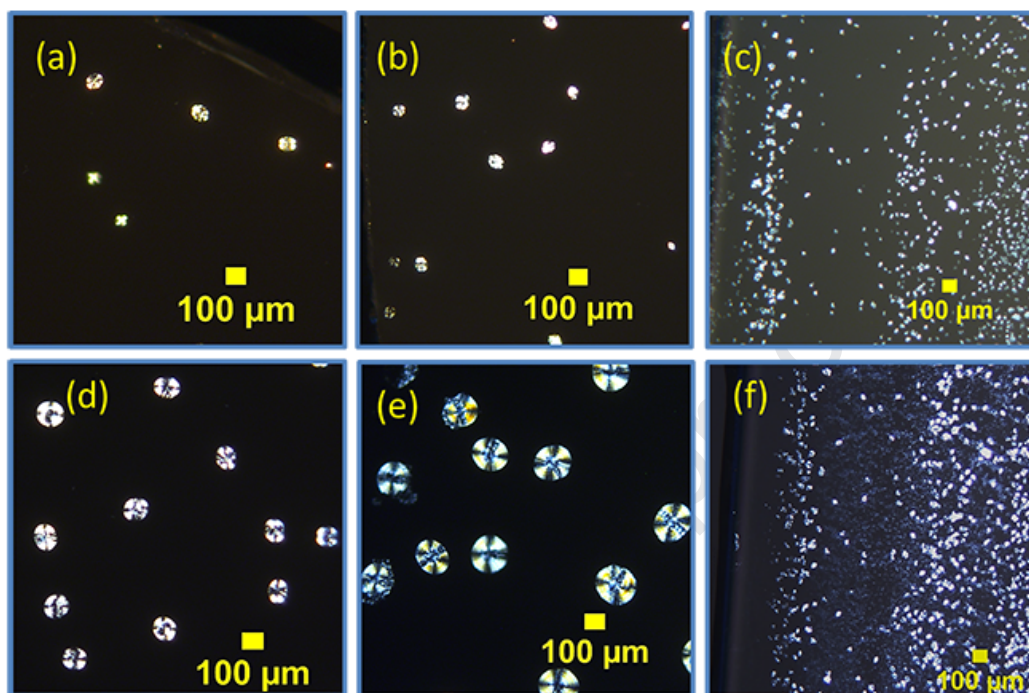


Figure 3: Representative POM images showing the effect of confinement on early stage growth of spherulite. The images in (a) and (d) correspond to 100 μm film; (b) and (e) are for 500 μm film and (c) and (f) correspond to 700 μm films. The images in top panel i.e., (a), (b) and (c) represent microstructure after 24 hours of incubation and those in bottom panel i.e., (d), (e) and (f) correspond to microstructure on 3rd day of incubation.

served. This characteristic feature of zone I in thicker films is observed within 24 hours of incubation, and no significant change in diameter of spherulites is observed after 3 days of incubation (Figure 3 (f) and Figure S 3 (ESI)). Further, zone II – the region of individual larger spherulites, is observed to be absent in the 700 μm and 850 μm films during the early stage (Table S1 (ESI)). As the film thickness decreases (increase in degree of confinement), the width of the zone I is observed to decrease, as shown in Figure S 2 (b),

during this stage. Moreover, in highly confined films (100 and 200 μm), only the growth of zone II is spotted in the early stage (Table S1 (ESI)). The absence of zone I in the thinner films is most likely due to the decreased nucleation caused by the smaller cross section area available for the diffusion of water molecules.

Furthermore, in thicker films (700 and 850 μm), as the growth of zone I proceeds, we observed the disappearance of spherulites close to the air-film interface i.e., appearance of zone depleted of spherulites, as seen in Figure 3(f). This region appears black under crossed polarization microscopy and thus devoid of any morphologies observable under microscope. This region resembles the skin-core structure reported in the case of shear induced crystallization [21–25]. More detailed discussion on this aspect is presented in the following sections.

Morphology of spherulites

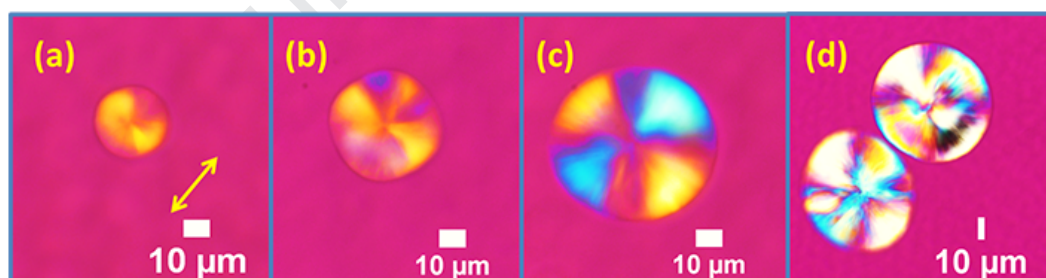


Figure 4: Different stages of growth of spherulites: The images represent the evolution of spherulitic morphologies after (a) 12, (b) 24, (c) 48 and (d) 72 hours of incubation in a 100 μm film. The yellow double headed arrow in (a) represents the orientation of z axis of tint plate.

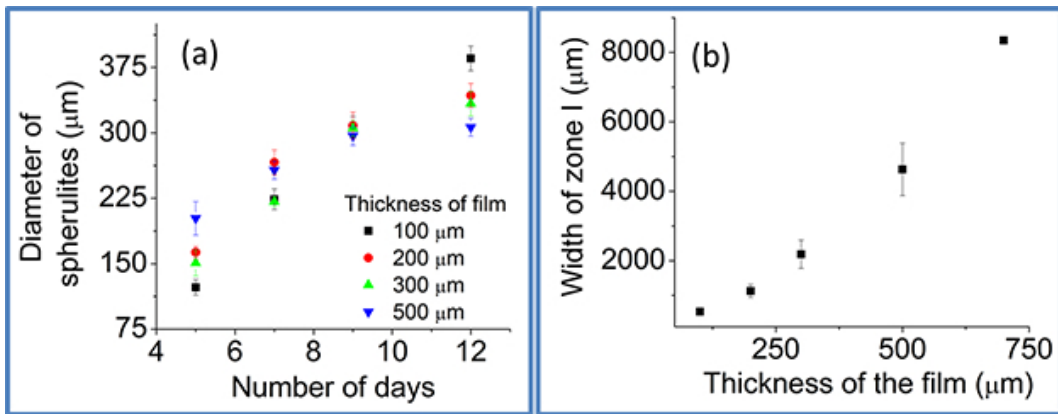


Figure 5: (a) Growth of spherulites with respect to film thickness in zone II and (b) span of zone I from the edge of the film towards the centre of the film with respect to the film thickness on 10th day of incubation.

Now we focus on the morphology of the spherulites formed during early growth stage. Figure 4 shows the growth stages of spherulites. Initial growth shown in Figure 4 (a) and (b) ultimately leads to the formation of positive type cellulose spherulites shown in Figure 4 (c), which has been reported earlier in unconfined cellulose/IL films [17]. However, the size of the spherulites was smaller than that observed in this study. This is probably due to the smaller area available for the diffusion of moisture, which in turn decreases the number of nuclei formation. But further growth of spherulites is found to deviate from the systematic positive type radial growth, as seen in Figure 4(d), resulting in the intermingling of different colors across the diameter.

3.2. Intermediate stage (3 – 10 days of incubation)

As the evolution of morphologies continues, apart from the change in the size of spherulites and the spanning of zone I further towards the film

interior, depending on the degree of confinement, we observed the evolution of interesting microstructural features in the zone I. As the growth proceeds, the size of spherulites increases with the increase in film thickness, except for films $\geq 700 \mu\text{m}$. In contrast to the growth pattern in the early stage, the diameter of the spherulites in films of intermediate thickness (for e.g., 300 and 500 μm) becomes smaller than the size of spherulites in the thinner films (100 and 200 μm), as shown in Figure 5 (a). Moreover, the zone II appears even in thicker films (700 and 850 μm) in the intermediate stage with the formation of $\sim 75 \mu\text{m}$ diameter spherulites (Figure S 4 (ESI)). Meanwhile, the zone I spans further into the film as the growth progresses (Figure 5 (b)), with the appearance of zone I even in the highly confined films (100 and 200 μm). From the above results, it is evident that the retarded growth of individual spherulites in thicker films is linked to the spanning of zone II. The growth of large individual spherulites is decelerated due to the increase in number of nuclei formed in the thicker films. Whereas, limited nucleation in the thinner films leads to slow re-organization of cellulose chains and the growth of the individual spherulites.

The next important aspect in the intermediate stage is the effect of confinement on the morphological variation observed in the zone I. Through-the-width morphological distribution for cellulose/IL film of different thickness is presented in Figure 6 and Figure 7. For confinement thickness $\geq 700 \mu\text{m}$, as evident from Figure 6, panel (a) and Figure S 4 (ESI), there are primarily two regions in zone I – marked as ‘A’ and ‘B’. Region ‘A’ of $\sim 350 \mu\text{m}$ width which is next to the air-film interface is devoid of any specific morphologies when observed under the microscope. This region is shown in Figure 6 ((a)-

(1)) with a double headed arrow as well as in Figure S 4 (ESI). The second region – ‘B’ shown in Figure 6 (a)-(2), (a)-(3) and (a)-(4), is composed of tiny spherulites grown in different planes. Different layers seen in Figure S 4 (ESI) in region ‘B’ show the propagation of densely nucleated region in different focal planes across the thickness of the film. The width of zone I in the film of thickness $\geq 700 \mu\text{m}$ is observed to be $\sim 5000 \mu\text{m}$. Beyond zone I, which is composed of the regions A and B, there exists zone II with larger spherulites. This particular morphological distribution – the regions A and B of zone I as well as the interior zone II – resembles the well known skin-core morphology, typically observed in the shear-induced crystallization [21–25]. The skin-core morphology observed in conventional shear-induced crystallization consists of a skin layer immediately adjacent to the applied shear field which is devoid of any significant morphologies when observed under polarized light, followed by a transition layer of superimposed spherulitic structures and a core layer with fully grown spherulitic morphologies. The skin layer is reported to be consisting of highly oriented polymer chains, even though such features are not observable under microscope [21, 25].

In the 300 and 500 μm films within 7 days of incubation, apart from the formation of skin layer and the zone I, a narrow ‘streak-like’ radial growth originating from the zone I (Figure 6, panel (b)) is observed. By comparing the images (b)-(2), (b)-(3) and (b)-(4) which represent the microstructure in the direction of propagation, it is clear that the streak-like growth intensifies as it propagates further into the film, although the width of the streak decreases. Towards the end of intermediate stage, the growth of smaller spherulites and the evolution of a complex morphology consisting of super-

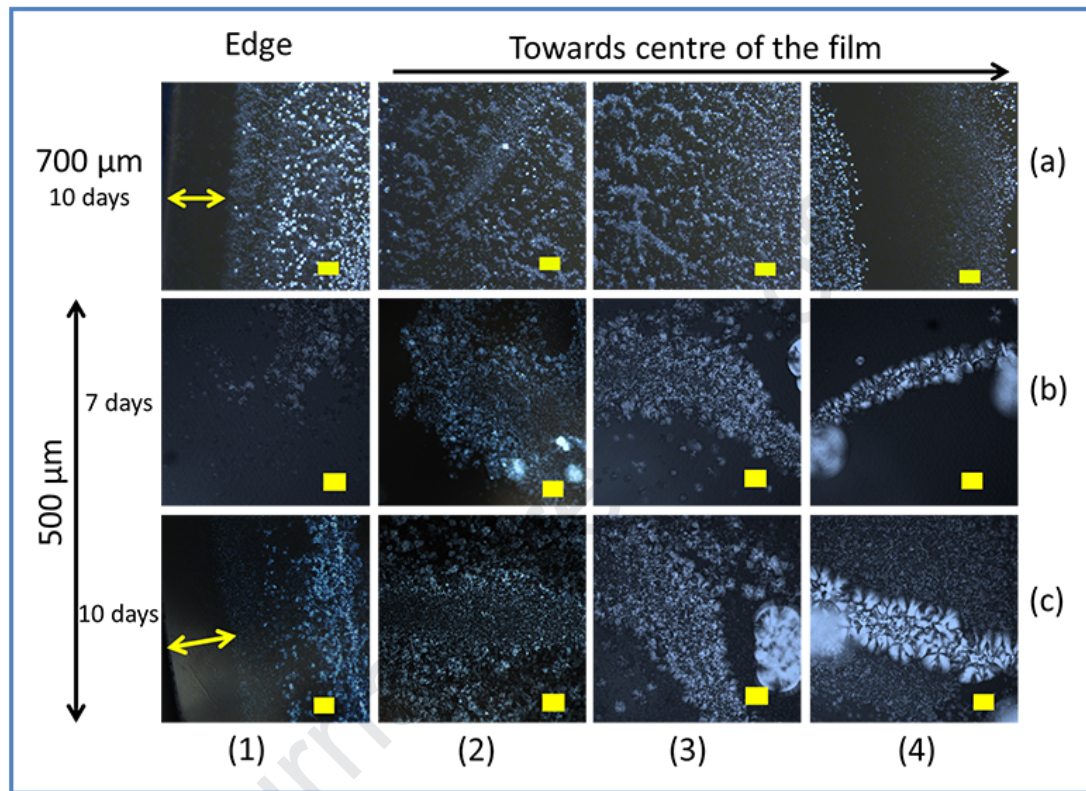


Figure 6: Propagation of microstructural changes in Zone I from the edge of the film towards the center. Panel (a) represents zone I in films of thickness $700\ \mu\text{m}$ on 10^{th} day and the panels (b) and (c) represent zone I in $500\ \mu\text{m}$ film, on 7^{th} and 10^{th} day, respectively. Panel (1) shows the microstructure at the edge and panels (2), (3) and (4) progressively towards the center and end of growth front of zone I. The yellow double sided arrows in (a)-(1) and (c)-(1) show the skin layer in the $700\ \mu\text{m}$ and $500\ \mu\text{m}$ films, respectively. Being in different planes, the nearby spherulites are seen out-of-focus in panel (b) and (c). A series of images corresponding to the images in panel (a) and (c) is stitched together to show the complete morphological features is given in ESI, Figure S 4 and S 5, respectively. The scale bars represent $100\ \mu\text{m}$.

imposed spherulitic structures or connected spherulites (similar to those in films of thickness $\geq 700 \mu\text{m}$) is observed to extend over the entire area of zone I along with the ‘streak-like’ growth, as seen in Figure 6, panel (c). Such complex superimposed structures have been observed in several other systems, associated with the higher nucleation rate [26, 27, 10].

Figure S 5 (ESI) shows through-the width distribution of streak-like growth and superimposed spherulitic morphology along the path of streak-like growth. Figure S 6 (ESI) gives a clear representative view of the skin layer and the origin of the streak-like morphology in 300 and 500 μm films. However, compared to the thicker films ($\geq 700 \mu\text{m}$), the area available for the diffusion of ambient moisture is less in this case. This leads to the streak like growth in zone I instead of the transition layer that forms in the thicker film. The fully grown spherulites (formed during the early stage) in this region act as nucleation sites for the formation of streak-like structure, as shown in Figure S 6 (b) (ESI). The radial growth of the streak-like morphology, when intercepted by the existing spherulites, bends around these spherulites as the growth propagates across the width of the film due to the unavailability of free cellulose chains in the vicinity of the existing spherulites. Further diffusion of moisture into the film causes the onset of fresh nucleation in different planes surrounding the streak-like structure, as shown in Figure 6, panel (c) and in Figure S 5 (ESI). As seen in Figure S 7 (ESI), the formation of superimposed spherulitic structure (marked as B) next to the streak (denoted as C), and the formation of small individual spherulites (A) away from the streak indicates that the streak acting as nucleating site for this fresh growth observed in zone I. This occurs towards the end of intermediate stage.

A detailed look at the radial streak-like structure (Figure S 8 (ESI)) shows the presence of two layers – an inner core layer of $\sim 70 \mu\text{m}$ (marked as B) consisting of densely populated and smaller superimposed spherulitic morphology and an outer layer (marked as A) of larger superimposed structures grown lateral to the inner core. This two layered morphology of streak-like growth is very similar to the cylindrite morphologies observed in shear-induced crystallization in terms of a densely nucleated core layer acting as nuclei for the lateral growth [28, 29, 21, 30, 31]. Unlike the cylindrites formed during shear-induced crystallization or in the presence of anisotropic nucleating agents, there is no driving force that enhances the stretching of polymer chains in the core region. In the present scenario, the evolution of the dense core layer is due to the diffusion of ambient moisture from the edge of the film to the interior. This causes the aggregation of cellulose molecules and hence nucleation along the direction of diffusion resulting in the formation of superimposed spherulitic structures. The outer layer of lateral overgrowth of lamellae is formed attached to the core region.

With further decrease in film thickness, the skin layer vanishes and the transition layer consisting of superimposed spherulitic structures (Figure 7 (b)) originates from the edge of the film, as seen in Figure 7, in 100 and 200 μm films.

Evolution of spherulites

Figure 8 shows different stages of growth observed during the secondary nucleation of tiny spherulitic structures in the zone I. The numbers, shown in Figure 8, in the ascending order represent different stages of development of a spherulitic morphology. Here we observe the initial stage as the rod-

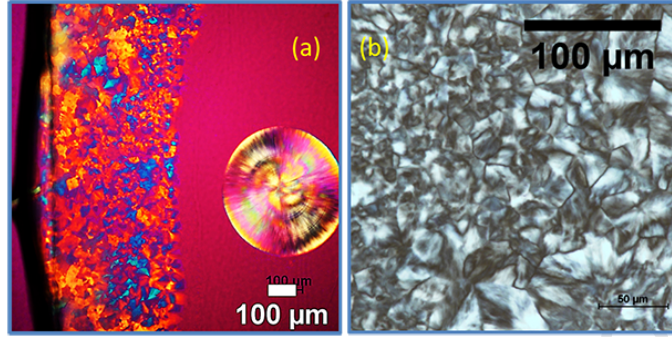


Figure 7: (a) Microstructural features of Zone I with superimposed spherulitic morphology and zone II with fully grown spherulite in a $100\ \mu\text{m}$ film on the 10^{th} day of incubation. The edge of the film can also be seen on the left side of the image. (b) Higher magnification image of the superimposed spherulitic structures.

like stacking of lamellae (marked as ‘1’) which splay to both sides to form a dumbbell shape (‘2’). However, after the formation of the dumbbell hinged at the central point during the initial stage, it bends to one side to form a small *arc* (‘3’), which further takes ‘C’ shape (‘4’). This C shaped object further grows to form a single *eye-like* region (‘5’), unlike the formation of *two-eyed* morphology reported earlier [10, 32–35]. We believe that this filled structure acts as a precursor for the formation of smaller spherulites (‘6’), which further grows into larger spherical or disk-like structure as shown in Figure 4 (a).

3.3. Terminal growth stage (11 - 25 days of incubation)

In the terminal growth stage, the size of the spherulites increases further as shown in Figure S 9. The diameter of the spherulites (zone II) in $\geq 700\ \mu\text{m}$ films increases to $\sim 300\ \mu\text{m}$ as the incubation proceeds for 25 days,

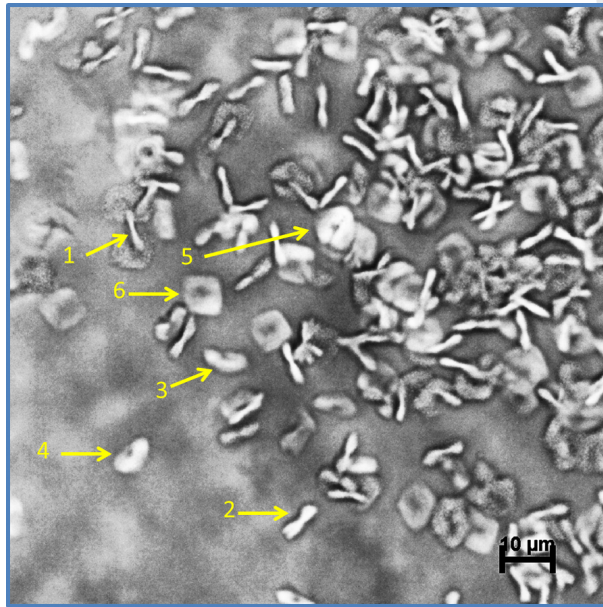


Figure 8: Spherulites at different stages of growth, marked in the ascending order from 1 – 6 observed during the fresh nucleation: (1) rod-like stacking of lamellae, (2) dumbbell shape, (3) *arc* shape, (4) filling the open side of ‘C’ with a single *eye-like* region, (5) growth of this structure with just one *eye* inside and (6) smaller spherulites, which further grow into larger spherical or disk-like structure as shown in Figure 4 (a).

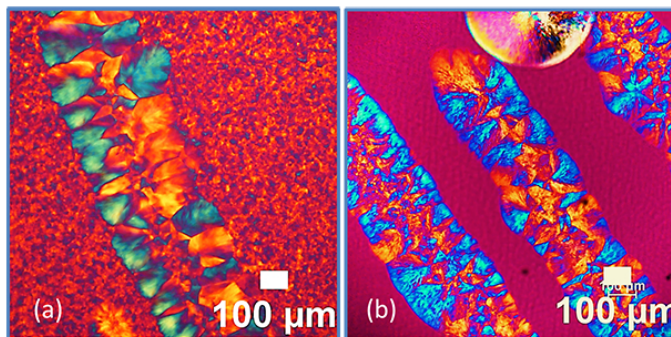


Figure 9: Representative images showing the growth of cylindrites in 500 μm films after 25 days of incubation period – (a) near the vicinity of air-film interface (in zone I within $\sim 2000 \mu\text{m}$ from the edge) and (b) in the interior of the film. Note that the cylindrites is surrounded by the superimposed spherulitic structures near the edge.

although significant morphological changes are not observed. Considerably larger spherulites are observed in zone II formed in $\leq 500 \mu\text{m}$ thick films. The size of the spherulites increases with distance from edge towards the center. As shown in Figure S 10 (ESI), an increase in size from ~ 250 to $400 \mu\text{m}$ is observed over a distance spanning $\sim 3000 \mu\text{m}$ from the air-film interface. Moreover, the size of the spherulite is observed to increase with the increase in confinement, as shown in Figure S 11 (ESI) most likely due to the difference in the net moisture that diffuses into the films. A higher standard deviation signifying a large difference in the diameter of spherulites is evident in all the films, as shown Figure S 9 (ESI). Moreover, we observed a systematic growth of zone I with increase in incubation period, which is consolidated as Figure S 12 (ESI), in different incubation periods (early, intermediate and terminal).

As the incubation period prolongs over 25 days, apart from the change

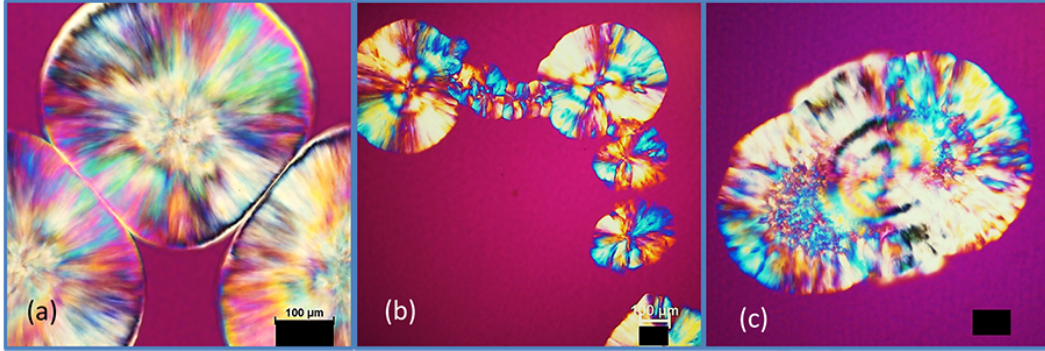


Figure 10: Morphology of spherulites formed in the zone I in 100 μm films: (a) interdigitation of large spherulites observed near the center of the film and (b) secondary nucleation bridging the gap between neighboring spherulites and (c) crystallization of cellulose around the primary spherulites leading to non-circular spherulitic structures observed near the zone I-zone II boundary. *Scale bars represent 100 μm .*

in the size of spherulites and growth of zones, growth of very distinct polycrystalline morphologies is observed in the confined films. In 300 – 500 μm films, as shown in Figure 9, the cylindrites propagate to the interior of the film for over 1000 μm from the edge of the film.

However, in the highly confined films considered (100 and 200 μm), the microstructure changes drastically with the distance from the center to the air-film interface. The interdigitation of larger spherulites is observed in the central part of the zone I (Figure 10 (a)). Moving towards edge, a bridge between the neighboring spherulites (Figure 10 (b)) appears with the primary spherulites acting as the nucleation centres for the bridge. Moreover, the individual spherulites which are located far apart grow in the direction of diffusion of moisture to form elongated spherulitic structures, as shown in Figure 10 (c). This deformation originates on a plane that is different

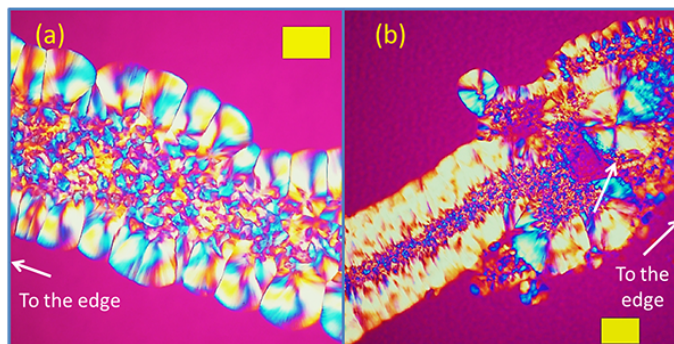


Figure 11: Presence of both (a) cylindrites and (b) *Shish-kebab* morphologies observed in $100\ \mu\text{m}$ films, nucleating directly from the zone I, near the edge of the film and from the primary spherulites (pointed by white arrow) near the edge. A magnified view of (b) is given in Figure S 14 (ESI). The arrows in both images shows the direction of air interface. Scale bars represent $100\ \mu\text{m}$.

from the plane of primary spherulites, as evident from Figure S 13 (ESI). If the individual spherulites are even farther, they grow into bigger, individual spherulites.

Moreover, near the air-film interface, we observed the presence of both cylindrites and *shish-kebab* morphologies originating from the transition layer of superimposed morphologies as well as from the primary spherulites located near the edge, as shown in Figure 11 (a) and (b). The coexistence of cylindrites and *shish-kebab* in the sub-skin region (the region between skin and transition layers) is previously observed in the shear induced crystallization. The formation of cylindrites and *shish-kebab* is reported to be a strong function of shear rate [28]. Even though the formation mechanism is same for both cylindrites and *shish-kebab*, the more number of row nuclei (leading to the formation of *shish* or the core) generated by high shear rates cause the

systematic lateral growth of lamellae attached to the *shish* [28].

We also observed the formation of *terraces* on the spherulites during prolonged incubation of the samples for over 3 months in the highly confined films (100 μm), as shown in Figure S 15 and S 16. These structures are reported earlier in case of crystallization under confined conditions, due to the instabilities during the growth of spherulites [36].

3.4. Avrami kinetics

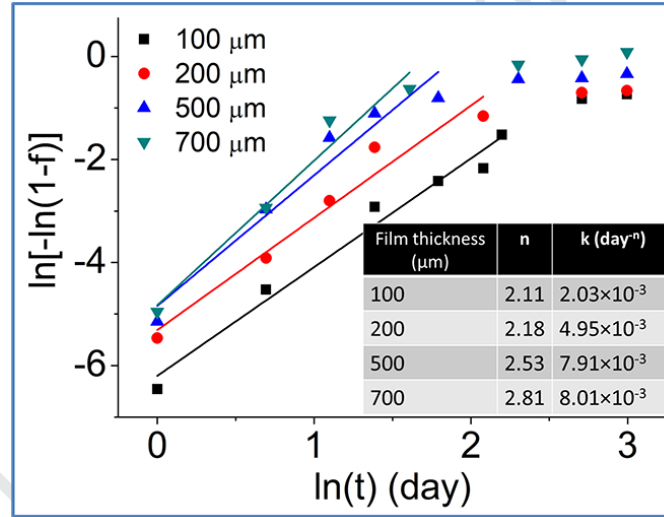


Figure 12: Avrami plots for the crystallization of cellulose at 40 °C at different confinement thickness as mentioned in the legend. The table in the inset shows the Avrami parameters (n and k).

In this section, we discuss the kinetics of cellulose crystallization. The classic Avrami kinetics [37] can be used to describe the crystallization of polymers, including crystallization under confinements [38–40]. According to the Avrami theory, the kinetics of crystallization process can be expressed

as:

$$f = 1 - \exp(-k.t^n) \quad (1)$$

where, f is the fraction crystallized, t is the time (duration), k is the crystallization rate constant and n is called the Avrami exponent, which describes the geometry of crystal growth and the nucleation mechanism. The equation 1 can be recast as:

$$\ln(-\ln(1 - f)) = \ln(k) + n.\ln(f) \quad (2)$$

which can then be fitted with a linear equation to obtain slope n and intercept $\ln(k)$. Figure 12 shows the Avrami plots at different film thickness when the MCC/AmimCl solution is incubated at 40 °C and 20% RH. Here, the fractional crystallization – f is estimated from the POM images as the ratio of the area occupied by the cellulose crystallized at a given time to the total area of the film. The plots of $\ln(-\ln(1-f))$ vs. $\ln(t)$ show a linear behavior in the initial growth period and deviate from linearity as the growth saturates. The data when fitted to the Avrami equation (equation 2) provide estimate of the Avrami parameters (n and k) which are tabulated in the inset of Figure 12. Even though, the rate constant, k showed only a marginal increase with increase in incubation period, the Avrami exponent, n increased from 2 to 3, with an increase in film thickness from 100 to 700 μm . Here, the value of n close to 2 in highly confined film (100 μm) indicates a two dimensional (2D) growth of polycrystalline morphologies [39, 40]. Moreover, n increases to ~ 3 , as the film thickness increases to 700 μm , signalling negligible effect of confinement. The formation of terraces on spherulites supports the strong

influence of confinement prevailing even in the 100 μm film, which is the most confined film considered in this study.

Furthermore, since the key parameter for the formation of morphologies in the present system is the diffusion of water from the edge to the interior of the film, the fraction of cellulose crystallized can be considered as a measure of weight fraction of water in the film at a given time instant. The ratio of weight of water present at time t to the weight of water at time $t = \infty$ ($\frac{M_t}{M_\infty}$) can be used to obtain the diffusivity of water, as reported earlier [41]. Figure S 17 (ESI), shows the plot of the fraction crystallized versus the incubation period as a function of thickness of film. This data, in principle, can be used to evaluate the diffusivity of water through the cellulose film with the aid of appropriate model, which is the work in progress.

4. Conclusions

Through-the width morphological distribution of polycrystalline cellulose structures formed due to the diffusion of the anti-solvent (ambient moisture) into MCC/AmimCl solution films confined between glass substrates is studied at various confinement thickness. The difference in the concentration of water that diffuses into the film results in the formation of zones with different morphologies – superimposed spherulitic structures, near the air-film interface and larger individual spherulites, in the interior of the film. Skin-transition-core layered morphological distribution composed of small spherulites ($\sim 50 \mu\text{m}$), similar to those found in shear-induced crystallization, is observed in the films of thickness $\geq 700 \mu\text{m}$. However, when the confinement thickness reduces ($\leq 500 \mu\text{m}$), cylindrites and *shish-kebab* mor-

phologies are formed. In the absence of shear to stretch the cellulose chains for the nucleation of the core region (*shish*), we hypothesize that the formation of core is caused by the diffusion of ambient moisture which lead to the nucleation of superimposed structures. The superimposed structure then acts as nucleating centres for the lateral growth of *kebabs*. Furthermore, the crystallization kinetics is observed to follow Avrami equation, with an Avrami exponent value ~ 2 in highly confined 100 μm films signalling a 2D growth of polycrystalline morphologies. In support of this, the formation of *terrace-like* concentric rings on the spherulites is observed in highly confined films after prolonged incubation, which further points to the strong influence of the confinement on the microstructural changes even in 100 μm thick film. The slow evolution of microstructure (over several weeks) offers a unique platform to use this as a model system for further studies on the morphological evolution of more industrially important structures such as, *shish-kebab* in the absence of shear and nanofillers, the skin-core distribution in shear-induced crystallization, etc.

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1. The growth of polycrystalline structures is studied in confined cellulose/ionic liquid films.
2. Crystallization of cellulose is induced via diffusion of moisture (anti-solvent).
3. Shish-kebab morphologies are observed to form due to directional diffusion of moisture.
4. Formation of terraces on spherulites is observed in highly confined films.
5. Growth kinetics followed Avrami equation at the confinement thicknesses considered.